

multiplicityFunction

September 25, 2026

0.1 import statements

```
[52]: %matplotlib widget
```

```
[53]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import scipy.special as sp
import math as math
```

```
[54]: pd.set_option('display.max_colwidth', None)
pd.set_option('display.max_columns', None)
pd.set_option('display.max_rows', None)
```

0.2 define some helpful functions

```
[55]: def factorial(x):
    return sp.factorial(x, exact=True)

def multiplicity(N,q):
    return factorial(q+N-1)//factorial(q)//factorial(N-1)

def logArray(array):
    import math as math
    return [math.log(x) for x in array]
```

0.3 Two Einstein Solids

0.3.1 3 particles each and 6 energy units

```
[56]: table1 = pd.DataFrame({'q_A':range(0, 6+1, 1), 'q_B':range(6,0-1,-1)})
table1['multi_A'] = [multiplicity(3,i) for i in table1['q_A']]
table1['multi_B'] = [multiplicity(3,i) for i in table1['q_B']]
table1['multi_total'] = table1['multi_A']*table1['multi_B']

table1
```

```
[56]:
```

	q_A	q_B	multi_A	multi_B	multi_total
0	0	6	1	28	28
1	1	5	3	21	63
2	2	4	6	15	90
3	3	3	10	10	100
4	4	2	15	6	90
5	5	1	21	3	63
6	6	0	28	1	28

how many total microstates from the above situation? First just add up the microstate from each macrostate

```
[57]: table1['multi_total'].sum()
```

```
[57]: 462
```

Alternatively, treat the two Einstein solids as one solid with the combined number of particles

```
[58]: multiplicity(6,6)
```

```
[58]: 462
```

most likely macrostate just sort the table for the macrostate with the highest total multiplicity.

```
[59]: table1.sort_values('multi_total', ascending=False).iloc[0]
```

```
[59]: q_A          3
      q_B          3
      multi_A      10
      multi_B      10
      multi_total  100
      Name: 3, dtype: int64
```

```
[60]: table1['probability'] = table1['multi_total']/table1['multi_total'].sum()
```

```
[61]: table1
```

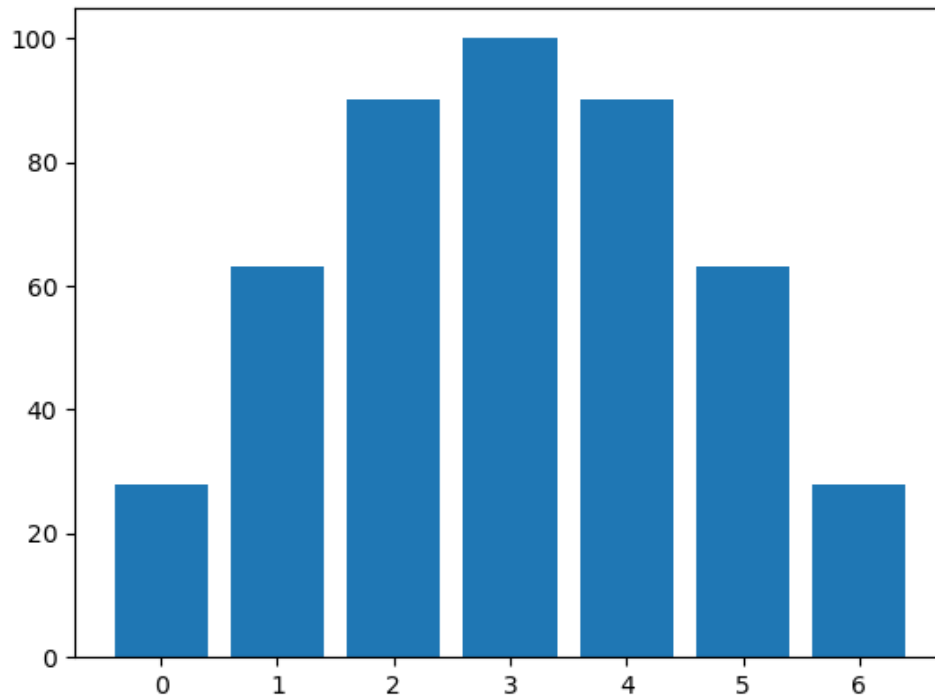
```
[61]:
```

	q_A	q_B	multi_A	multi_B	multi_total	probability
0	0	6	1	28	28	0.060606
1	1	5	3	21	63	0.136364
2	2	4	6	15	90	0.194805
3	3	3	10	10	100	0.216450
4	4	2	15	6	90	0.194805
5	5	1	21	3	63	0.136364
6	6	0	28	1	28	0.060606

```
[62]: fig1 = plt.figure()
      ax1 = fig1.add_subplot()
```

```
ax1.bar(table1['q_A'],table1['multi_total'])
```

[62]: <BarContainer object of 7 artists>



```
[63]: def multiTable(N_a, N_b, q):
    df = pd.DataFrame({'q_A':range(0, q+1, 1),'q_B':range(q, 0-1,-1)})
    df['multi_A'] = [multiplicity(N_a, i) for i in df['q_A']]
    df['multi_B'] = [multiplicity(N_b, i) for i in df['q_B']]
    df['multi_total'] = df['multi_A']*df['multi_B']
    df['probability'] = df['multi_total']/df['multi_total'].sum()
    return df

def multiLargeN(N_a, N_b, q):
    q_A = np.arange(0, q+1, 1)
    q_B = np.arange(q, 0-1, -1)
    multi_A = np.array([multiplicity(N_a, i) for i in q_A])
    multi_B = np.array([multiplicity(N_b, i) for i in q_B])
    multi_total = multi_A*multi_B
```

```

probability = multi_total/multi_total.sum()
return q_A, probability

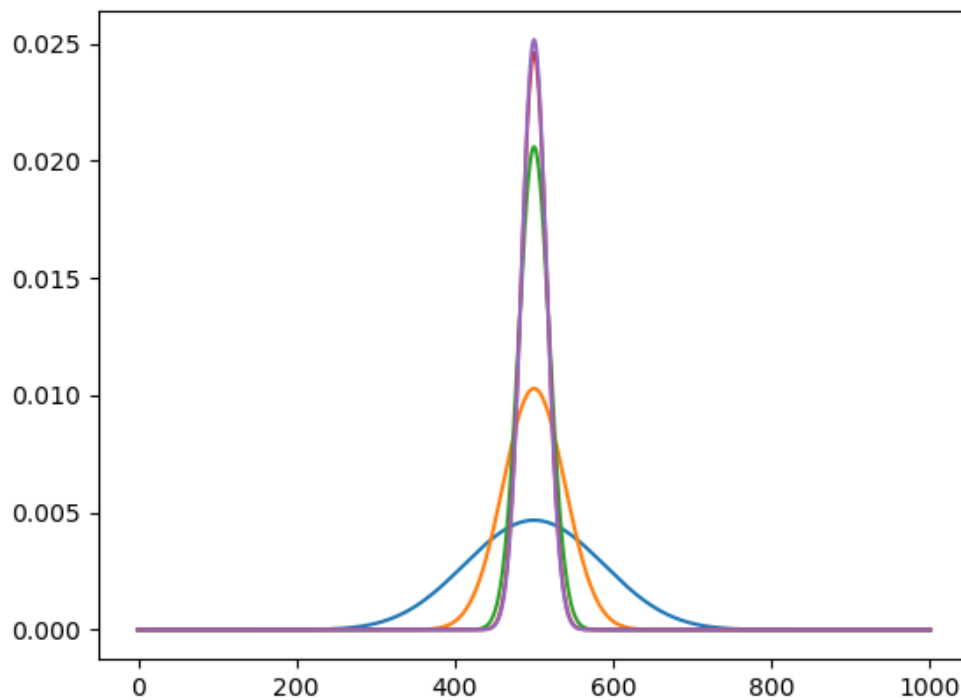
```

```

[64]: fig0 = plt.figure()
      ax0 = fig0.add_subplot(111)

      for i in [18, 100, 1000, 10000, 100000]:
          q_A, probability = multiLargeN(i,i,1000)
          ax0.plot(q_A, probability)

```



```
[ ]:
```

```
[ ]:
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```

[65]: fig0.savefig('multipliticitySharpness-1.png', dpi=300)

```

```

[66]: ax0.set_xlim(300,500)
      ax0.set_ylim(0,0.005)

```

```
[66]: (0.0, 0.005)
```

```
[67]: fig0.savefig('multiplitcitySharpness-2.png', dpi=300)
```

```
[68]: def multiplicityApprox(N,q):
    return (((q+N)/q)**q)*(((q+N)/N)**N)/np.sqrt(2*np.pi*q*(q+N)/N)

def multiEvenLargerN(N_a, N_b, q):
    q_A = np.arange(1, q+1, 1)
    q_B = np.arange(q, 1-1, -1)
    multi_A = np.array([multiplicityApprox(N_a, i) for i in q_A], dtype=np.
↪float64)
    multi_B = np.array([multiplicityApprox(N_b, i) for i in q_B], dtype=np.
↪float64)
    multi_total = multi_A*multi_B
    probability = multi_total/multi_total.sum()
    return q_A, probability
```

0.4 got this from chatgpt

```
[71]: from mpmath import mp

# mp.dps = 50 # set decimal places of precision

def multiplicityApprox_mp(N, q):
    N = mp.mpf(N)
    q = mp.mpf(q)
    return (((q + N) / q) ** q) * (((q + N) / N) ** N) / mp.sqrt(2 * mp.pi * q_
↪* (q + N) / N)

def multiEvenLargerN_mp(N_a, N_b, q):
    q_A = [mp.mpf(i) for i in range(1, q + 1)]
    q_B = [mp.mpf(i) for i in range(q, 0, -1)]

    multi_A = [multiplicityApprox_mp(N_a, i) for i in q_A]
    multi_B = [multiplicityApprox_mp(N_b, i) for i in q_B]

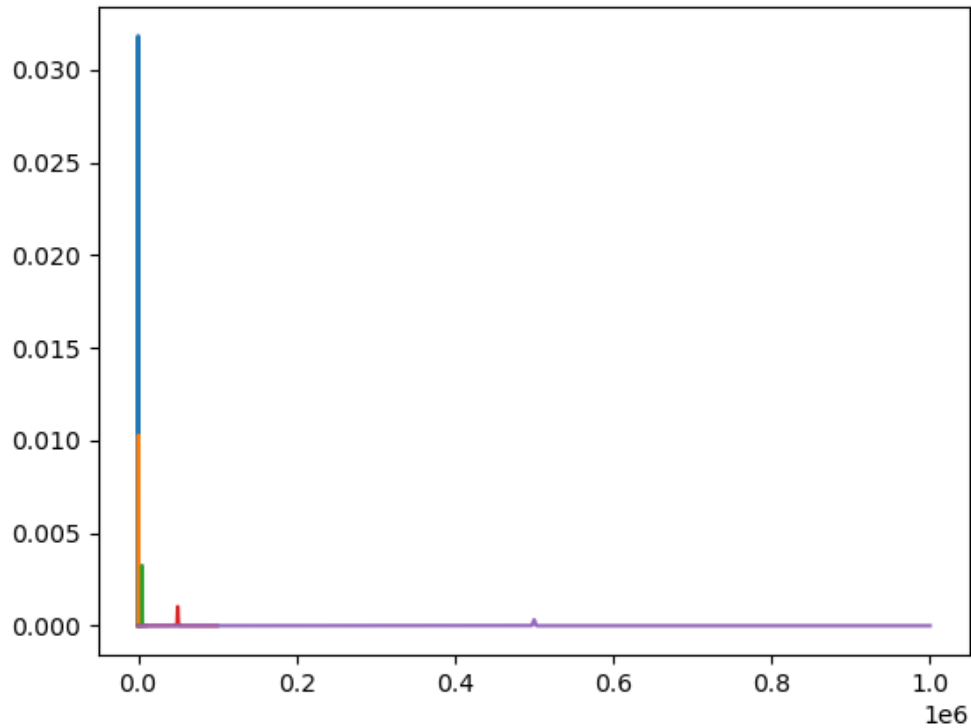
    multi_total = [a * b for a, b in zip(multi_A, multi_B)]
    total_sum = sum(multi_total)
    probability = [m / total_sum for m in multi_total]

    # convert to float for plotting
    probability_float = [float(p) for p in probability]

    return [float(i) for i in q_A], probability_float
```

```
[72]: fig2 = plt.figure()
ax2 = fig2.add_subplot(111)
```

```
for i in [10, 100, 1000, 10000, 100000]:
    q_A, probability = multiEvenLargerN_mp(i,i,10*i)
    ax2.plot(q_A, probability)
```



```
[48]: fig2.savefig('multiplicity-fig2.png', dpi=300)
```

```
[ ]:
```

```
[ ]:
```

Make these numbers floats instead of integers so we can read them better. Note that this will break for numbers much larger than this.

```
[13]: table2 = multiTable(300, 200, 100)
```

```
[14]: table2
```

```
[14]:
```

	q_A	q_B	\
0	0	100	
1	1	99	
2	2	98	

3	3	97
4	4	96
5	5	95
6	6	94
7	7	93
8	8	92
9	9	91
10	10	90
11	11	89
12	12	88
13	13	87
14	14	86
15	15	85
16	16	84
17	17	83
18	18	82
19	19	81
20	20	80
21	21	79
22	22	78
23	23	77
24	24	76
25	25	75
26	26	74
27	27	73
28	28	72
29	29	71
30	30	70
31	31	69
32	32	68
33	33	67
34	34	66
35	35	65
36	36	64
37	37	63
38	38	62
39	39	61
40	40	60
41	41	59
42	42	58
43	43	57
44	44	56
45	45	55
46	46	54
47	47	53
48	48	52
49	49	51

50	50	50
51	51	49
52	52	48
53	53	47
54	54	46
55	55	45
56	56	44
57	57	43
58	58	42
59	59	41
60	60	40
61	61	39
62	62	38
63	63	37
64	64	36
65	65	35
66	66	34
67	67	33
68	68	32
69	69	31
70	70	30
71	71	29
72	72	28
73	73	27
74	74	26
75	75	25
76	76	24
77	77	23
78	78	22
79	79	21
80	80	20
81	81	19
82	82	18
83	83	17
84	84	16
85	85	15
86	86	14
87	87	13
88	88	12
89	89	11
90	90	10
91	91	9
92	92	8
93	93	7
94	94	6
95	95	5
96	96	4

97	97	3
98	98	2
99	99	1
100	100	0

```

multi_A \
0
1
1
300
2
45150
3
4545100
4
344291325
5
20932912560
6
1064089721800
7
46515922124400
8
1785048511523850
9
61088326838816200
10
1887629299319420580
11
53196825708092761800
12
1378684399601404076650
13
33088425590433697839600
14
739762657843267673128200
15
15485698304185736624150320
16
304874685363656689787959425
17
5667082386759736116058539900
18
99803617589046463821697619350
19
1670397389121935552384202260700
20

```

26642838356494872060528026058165
21
405986108289445669493760397076800
22
5923706398223275450340776702802400
23
82931889575125856304770873839233600
24
1116125013865235482768374677086352200
25
14464980179693451856678135815039124512
26
180812252246168148208476697687989056400
27
2183140527120400604294940868380904903200
28
25495962584584678485873059427162710833800
29
288368128542888777357460810072736867361600
30
3162437143020346925020153550464347645398880
31
33664653457958531782472602311394668483278400
32
348218759205758563124950980158488602123910950
33
3503291759282177059317688648867218663792073800
34
34311651642381322375082068237434818089492958100
35
327431189958724619236497451180092264054018514440
36
3046929128782576317895184615148080790502672287150
37
27669410466782314670615730559182571502943186175200
38
245383982297516843263092136801171752539259308974800
39
2126661179911812641613465185610155188673580677781600
40
18023453499752612137674117448046065224008596244199060
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149462785119899710409980486154528345760071285927504400
42
1213495469663947648804841566159384902480578773839976200
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44
75238001883931840851321746701846091633396814822448334100
45
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46
4313645441345425542142446810905842586981417383153704488400
47
31755772823521643352793331841987692236075966267471952191200
48
229567774370041880071235128107702691789965839475265987715550
49
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50
11380189931751153640168019909770084621744869851408165200011828
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527170563014943146566606804643761272919063823999054711471136150
53
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5832502475076523506054565947487114238462258603009368833794280170400
58
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217834252288027517011925732169648779182030222973762534532995842039200
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3573306861400455562000
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3937649830102201883000
79
7181220012330924929686122460182810216751686499665460644469260
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0439919976623546633250
83
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84
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4515624225456771905375
85
70303057592233220427747303008840737855453149732419197601857131704
2517643559710593078400
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8713484548577621656000
95 154040253319087246516493261587317418215250482810217368344893045750931425
8032767496206136131200
96 633811458969161066395987899239483127031499382396206880169091177829353262
4197324593764830956500
97 2587518945894719405080527918544694003138904695143277572649073262066225689
8784954011658485142000
98 10482092056328608202213975343492280808634134326243685676955939643268281621
2424762679881822463000
99 42140127660795818833143052391009371331680661230757443428570343212331071568
2273288349423892326000
100 168139109366575317144240779040127391613405838310722199279995669417200975557
2270420514201330380740

multi_B \

0 277216764217237649652225568421760372018697733716243247244202243820317088554
9339080
1 92714636861952391188035307164468351845718305590716805098395399271009059717
3692000
2 30801171306487539354414414125108613532637960582150884915238740026274821852
4146000
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0964000
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96159170546680515070004296638162160957776561405035514504559985005098616800
16
28780033438267055566726638078323181976799322955732460327069009596596417000
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34998117046665338647557789430657855944726546218817762765294783000
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35
559156732379182037666034763113055835005974419928006681558309600

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8055759120568294423909647151975906199308729765733334955152000
39
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448966710786756403053732471688371312198432301414728452702400
41
104007732228592216923644588035916134100022926968662962016000
42
23784713959251708521298568581856790356206793376554708368000
43
5367756457729957565117964893959898212684801618055148192000
44
1195164523791435864108296870920758586418100360270091589625
45
262467503264001601529665195182597964076131843824020113800
46
56833514486299559386344825728515307181839572481579158500
47
12130473447668680659536049760236468726558643928874603000
48
2551250367962063789505597767033860486141302096152198250
49
528545892964252259180442565281915319838038681274559000
50
107823362164707460872810283317510725246959890980010036
51
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5	2.44964e-27
6	4.02371e-26
7	5.64301e-25
8	6.89698e-24
9	7.46212e-23
10	7.23543e-22
11	6.35006e-21
12	5.08574e-20
13	3.74254e-19
14	2.54528e-18
15	1.60778e-17
16	9.47369e-17
17	5.22697e-16
18	2.70935e-15
19	1.32327e-14
20	6.10569e-14
21	2.66779e-13
22	1.10615e-12
23	4.36072e-12
24	1.63731e-11
25	5.86432e-11
26	2.0065e-10
27	6.56691e-10
28	2.05828e-09
29	6.18506e-09
30	1.78366e-08
31	4.94096e-08
32	1.31584e-07
33	3.37152e-07
34	8.31732e-07
35	1.97679e-06
36	4.5291e-06
37	1.00086e-05
38	2.13432e-05
39	4.39402e-05

40	8.73693e-05
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43	0.000559365
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52	0.0243492
53	0.0314265
54	0.0392501
55	0.0474321
56	0.0554544
57	0.0627129
58	0.0685884
59	0.0725293
60	0.0741361
61	0.0732258
62	0.0698663
63	0.0643681
64	0.0572384
65	0.0491032
66	0.0406173
67	0.0323773
68	0.0248555
69	0.0183637
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81	1.70295e-05
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[15]: table2 = table2.astype(float)
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[16]: table2
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[16]:
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2	2.0	98.0	4.515000e+04	3.080117e+80	1.390673e+85	1.501521e-31
3	3.0	97.0	4.545100e+06	1.016335e+80	4.619344e+86	4.987544e-30
4	4.0	96.0	3.442913e+08	3.330557e+79	1.146682e+88	1.238082e-28
5	5.0	95.0	2.093291e+10	1.083842e+79	2.268798e+89	2.449640e-27
6	6.0	94.0	1.064090e+12	3.502212e+78	3.726667e+90	4.023714e-26
7	7.0	93.0	4.651592e+13	1.123576e+78	5.226419e+91	5.643009e-25
8	8.0	92.0	1.785049e+15	3.578514e+77	6.387821e+92	6.896984e-24
9	9.0	91.0	6.108833e+16	1.131351e+77	6.911237e+93	7.462120e-23
10	10.0	90.0	1.887629e+18	3.550103e+76	6.701278e+94	7.235426e-22
11	11.0	89.0	5.319683e+19	1.105568e+76	5.881273e+95	6.350059e-21
12	12.0	88.0	1.378684e+21	3.416513e+75	4.710294e+96	5.085744e-20
13	13.0	87.0	3.308843e+22	1.047572e+75	3.466251e+97	3.742540e-19
14	14.0	86.0	7.397627e+23	3.186670e+74	2.357380e+98	2.545283e-18
15	15.0	85.0	1.548570e+25	9.615917e+73	1.489092e+99	1.607785e-17
16	16.0	84.0	3.048747e+26	2.878003e+73	8.774304e+99	9.473689e-17
17	17.0	83.0	5.667082e+27	8.542483e+72	4.841096e+100	5.226972e-16
18	18.0	82.0	9.980362e+28	2.514277e+72	2.509339e+101	2.709355e-15
19	19.0	81.0	1.670397e+30	7.337036e+71	1.225577e+102	1.323265e-14
20	20.0	80.0	2.664284e+31	2.122500e+71	5.654942e+102	6.105688e-14
21	21.0	79.0	4.059861e+32	6.086021e+70	2.470840e+103	2.667787e-13
22	22.0	78.0	5.923706e+33	1.729481e+70	1.024494e+104	1.106154e-12
23	23.0	77.0	8.293189e+34	4.870018e+69	4.038798e+104	4.360724e-12
24	24.0	76.0	1.116125e+36	1.358664e+69	1.516439e+105	1.637312e-11
25	25.0	75.0	1.446498e+37	3.754854e+68	5.431389e+105	5.864317e-11
26	26.0	74.0	1.808123e+38	1.027789e+68	1.858368e+106	2.006495e-10

27	27.0	73.0	2.183141e+39	2.785947e+67	6.082114e+106	6.566910e-10
28	28.0	72.0	2.549596e+40	7.476991e+66	1.906331e+107	2.058281e-09
29	29.0	71.0	2.883681e+41	1.986507e+66	5.728452e+107	6.185058e-09
30	30.0	70.0	3.162437e+42	5.223777e+65	1.651987e+108	1.783664e-08
31	31.0	69.0	3.366465e+43	1.359347e+65	4.576195e+108	4.940956e-08
32	32.0	68.0	3.482188e+44	3.499812e+64	1.218700e+109	1.315841e-07
33	33.0	67.0	3.503292e+45	8.913378e+63	3.122616e+109	3.371515e-07
34	34.0	66.0	3.431165e+46	2.245099e+63	7.703305e+109	8.317323e-07
35	35.0	65.0	3.274312e+47	5.591567e+62	1.830854e+110	1.976788e-06
36	36.0	64.0	3.046929e+48	1.376712e+62	4.194743e+110	4.529099e-06
37	37.0	63.0	2.766941e+49	3.350173e+61	9.269731e+110	1.000861e-05
38	38.0	62.0	2.453840e+50	8.055759e+60	1.976754e+111	2.134318e-05
39	39.0	61.0	2.126661e+51	1.913629e+60	4.069640e+111	4.394024e-05
40	40.0	60.0	1.802345e+52	4.489667e+59	8.091931e+111	8.736925e-05
41	41.0	59.0	1.494628e+53	1.040077e+59	1.554529e+112	1.678437e-04
42	42.0	58.0	1.213495e+54	2.378471e+58	2.886264e+112	3.116324e-04
43	43.0	57.0	9.651522e+54	5.367756e+57	5.180702e+112	5.593647e-04
44	44.0	56.0	7.523800e+55	1.195165e+57	8.992179e+112	9.708931e-04
45	45.0	55.0	5.751527e+56	2.624675e+56	1.509589e+113	1.629916e-03
46	46.0	54.0	4.313645e+57	5.683351e+55	2.451596e+113	2.647009e-03
47	47.0	53.0	3.175577e+58	1.213047e+55	3.852126e+113	4.159172e-03
48	48.0	52.0	2.295678e+59	2.551250e+54	5.856849e+113	6.323689e-03
49	49.0	51.0	1.630400e+60	5.285459e+53	8.617411e+113	9.304290e-03
50	50.0	50.0	1.138019e+61	1.078234e+53	1.227050e+114	1.324857e-02
51	51.0	49.0	7.809934e+61	2.165128e+52	1.690951e+114	1.825733e-02
52	52.0	48.0	5.271706e+62	4.277873e+51	2.255169e+114	2.434925e-02
53	53.0	47.0	3.501208e+63	8.313276e+50	2.910651e+114	3.142654e-02
54	54.0	46.0	2.288753e+64	1.588309e+50	3.635246e+114	3.925006e-02
55	55.0	45.0	1.473125e+65	2.982131e+49	4.393050e+114	4.743213e-02
56	56.0	44.0	9.338557e+65	5.499832e+48	5.136049e+114	5.545435e-02
57	57.0	43.0	5.832502e+66	9.958543e+47	5.808323e+114	6.271295e-02
58	58.0	42.0	3.590006e+67	1.769493e+47	6.352491e+114	6.858837e-02
59	59.0	41.0	2.178343e+68	3.083764e+46	6.717494e+114	7.252935e-02
60	60.0	40.0	1.303375e+69	5.268097e+45	6.866305e+114	7.413607e-02
61	61.0	39.0	7.692049e+69	8.816899e+44	6.782001e+114	7.322584e-02
62	62.0	38.0	4.478757e+70	1.444786e+44	6.470846e+114	6.986626e-02
63	63.0	37.0	2.573508e+71	2.316534e+43	5.961620e+114	6.436811e-02
64	64.0	36.0	1.459662e+72	3.631855e+42	5.301279e+114	5.723836e-02
65	65.0	35.0	8.174106e+72	5.563692e+41	4.547821e+114	4.910320e-02
66	66.0	34.0	4.520528e+73	8.321762e+40	3.761876e+114	4.061729e-02
67	67.0	33.0	2.469423e+74	1.214334e+40	2.998705e+114	3.237727e-02
68	68.0	32.0	1.332762e+75	1.727286e+39	2.302061e+114	2.485555e-02
69	69.0	31.0	7.108064e+75	2.392777e+38	1.700801e+114	1.836369e-02
70	70.0	30.0	3.746965e+76	3.225047e+37	1.208414e+114	1.304735e-02
71	71.0	29.0	1.952644e+77	4.224953e+36	8.249828e+113	8.907409e-03
72	72.0	28.0	1.006154e+78	5.373844e+35	5.406914e+113	5.837890e-03
73	73.0	27.0	5.127250e+78	6.628530e+34	3.398613e+113	3.669511e-03

74	74.0	26.0	2.584411e+79	7.919040e+33	2.046606e+113	2.209737e-03
75	75.0	25.0	1.288760e+80	9.150890e+32	1.179330e+113	1.273332e-03
76	76.0	24.0	6.359012e+80	1.021305e+32	6.494489e+112	7.012154e-04
77	77.0	23.0	3.105180e+81	1.099162e+31	3.413096e+112	3.685148e-04
78	78.0	22.0	1.500837e+82	1.138771e+30	1.709110e+112	1.845341e-04
79	79.0	21.0	7.181220e+82	1.133619e+29	8.140765e+111	8.789652e-05
80	80.0	20.0	3.402103e+83	1.082091e+28	3.681383e+111	3.974821e-05
81	81.0	19.0	1.596048e+84	9.882105e+26	1.577232e+111	1.702950e-05
82	82.0	18.0	7.415785e+84	8.612844e+25	6.387100e+110	6.896206e-06
83	83.0	17.0	3.413048e+85	7.144295e+24	2.438382e+110	2.632742e-06
84	84.0	16.0	1.556187e+86	5.622825e+23	8.750169e+109	9.447631e-07
85	85.0	15.0	7.030306e+86	4.184428e+22	2.941781e+109	3.176265e-07
86	86.0	14.0	3.147288e+87	2.933010e+21	9.231027e+108	9.966817e-08
87	87.0	13.0	1.396383e+88	1.927800e+20	2.691947e+108	2.906518e-08
88	88.0	12.0	6.140911e+88	1.182141e+19	7.259426e+107	7.838063e-09
89	89.0	11.0	2.677161e+89	6.723080e+17	1.799877e+107	1.943342e-09
90	90.0	10.0	1.157129e+90	3.521613e+16	4.074959e+106	4.399768e-10
91	91.0	9.0	4.959123e+90	1.684982e+15	8.356034e+105	9.022080e-11
92	92.0	8.0	2.107627e+91	7.290789e+13	1.536627e+105	1.659109e-11
93	93.0	7.0	8.883762e+91	2.817696e+12	2.503174e+104	2.702698e-12
94	94.0	6.0	3.714169e+92	9.574696e+10	3.556203e+103	3.839663e-13
95	95.0	5.0	1.540403e+93	2.802350e+09	4.316747e+102	4.660828e-14
96	96.0	4.0	6.338115e+93	6.868505e+07	4.353337e+101	4.700335e-15
97	97.0	3.0	2.587519e+94	1.353400e+06	3.501948e+100	3.781083e-16
98	98.0	2.0	1.048209e+95	2.010000e+04	2.106901e+99	2.274838e-17
99	99.0	1.0	4.214013e+95	2.000000e+02	8.428026e+97	9.099810e-19
100	100.0	0.0	1.681391e+96	1.000000e+00	1.681391e+96	1.815412e-20

Total multiplicity of all macrostates:

```
[17]: total = table2['multi_total'].sum()
total
```

```
[17]: 9.261760158884879e+115
```

Total multiplicity of all states as a combined Einstein solid.

```
[18]: float(multiplicity(500,100))
```

```
[18]: 9.261760158884879e+115
```

Most probable macrostate

```
[19]: table2.sort_values('multi_total', ascending=False).iloc[0]
```

```
[19]: q_A          6.000000e+01
q_B          4.000000e+01
multi_A      1.303375e+69
multi_B      5.268097e+45
```

```
multi_total    6.866305e+114
probability    7.413607e-02
Name: 60, dtype: float64
```

Multiplicity of the most common macrostate:

```
[20]: multi_max = table2['multi_total'].sort_values(ascending=False).iloc[0]
multi_max
```

```
[20]: 6.866305444480905e+114
```

```
[21]: table2[55:65]
```

```
[21]:
```

	q_A	q_B	multi_A	multi_B	multi_total	probability
55	55.0	45.0	1.473125e+65	2.982131e+49	4.393050e+114	0.047432
56	56.0	44.0	9.338557e+65	5.499832e+48	5.136049e+114	0.055454
57	57.0	43.0	5.832502e+66	9.958543e+47	5.808323e+114	0.062713
58	58.0	42.0	3.590006e+67	1.769493e+47	6.352491e+114	0.068588
59	59.0	41.0	2.178343e+68	3.083764e+46	6.717494e+114	0.072529
60	60.0	40.0	1.303375e+69	5.268097e+45	6.866305e+114	0.074136
61	61.0	39.0	7.692049e+69	8.816899e+44	6.782001e+114	0.073226
62	62.0	38.0	4.478757e+70	1.444786e+44	6.470846e+114	0.069866
63	63.0	37.0	2.573508e+71	2.316534e+43	5.961620e+114	0.064368
64	64.0	36.0	1.459662e+72	3.631855e+42	5.301279e+114	0.057238

0.4.1 Now to add in the Entropy

```
[23]: table2['S_a'] = logArray(table2['multi_A'])
table2['S_b'] = logArray(table2['multi_B'])
table2['S_total'] = logArray(table2['multi_total'])
```

```
[24]: table2
```

```
[24]:
```

	q_A	q_B	multi_A	multi_B	multi_total	probability	\
0	0.0	100.0	1.000000e+00	2.772168e+81	2.772168e+81	2.993133e-35	
1	1.0	99.0	3.000000e+02	9.271464e+80	2.781439e+83	3.003143e-33	
2	2.0	98.0	4.515000e+04	3.080117e+80	1.390673e+85	1.501521e-31	
3	3.0	97.0	4.545100e+06	1.016335e+80	4.619344e+86	4.987544e-30	
4	4.0	96.0	3.442913e+08	3.330557e+79	1.146682e+88	1.238082e-28	
5	5.0	95.0	2.093291e+10	1.083842e+79	2.268798e+89	2.449640e-27	
6	6.0	94.0	1.064090e+12	3.502212e+78	3.726667e+90	4.023714e-26	
7	7.0	93.0	4.651592e+13	1.123576e+78	5.226419e+91	5.643009e-25	
8	8.0	92.0	1.785049e+15	3.578514e+77	6.387821e+92	6.896984e-24	
9	9.0	91.0	6.108833e+16	1.131351e+77	6.911237e+93	7.462120e-23	
10	10.0	90.0	1.887629e+18	3.550103e+76	6.701278e+94	7.235426e-22	
11	11.0	89.0	5.319683e+19	1.105568e+76	5.881273e+95	6.350059e-21	
12	12.0	88.0	1.378684e+21	3.416513e+75	4.710294e+96	5.085744e-20	
13	13.0	87.0	3.308843e+22	1.047572e+75	3.466251e+97	3.742540e-19	

14	14.0	86.0	7.397627e+23	3.186670e+74	2.357380e+98	2.545283e-18
15	15.0	85.0	1.548570e+25	9.615917e+73	1.489092e+99	1.607785e-17
16	16.0	84.0	3.048747e+26	2.878003e+73	8.774304e+99	9.473689e-17
17	17.0	83.0	5.667082e+27	8.542483e+72	4.841096e+100	5.226972e-16
18	18.0	82.0	9.980362e+28	2.514277e+72	2.509339e+101	2.709355e-15
19	19.0	81.0	1.670397e+30	7.337036e+71	1.225577e+102	1.323265e-14
20	20.0	80.0	2.664284e+31	2.122500e+71	5.654942e+102	6.105688e-14
21	21.0	79.0	4.059861e+32	6.086021e+70	2.470840e+103	2.667787e-13
22	22.0	78.0	5.923706e+33	1.729481e+70	1.024494e+104	1.106154e-12
23	23.0	77.0	8.293189e+34	4.870018e+69	4.038798e+104	4.360724e-12
24	24.0	76.0	1.116125e+36	1.358664e+69	1.516439e+105	1.637312e-11
25	25.0	75.0	1.446498e+37	3.754854e+68	5.431389e+105	5.864317e-11
26	26.0	74.0	1.808123e+38	1.027789e+68	1.858368e+106	2.006495e-10
27	27.0	73.0	2.183141e+39	2.785947e+67	6.082114e+106	6.566910e-10
28	28.0	72.0	2.549596e+40	7.476991e+66	1.906331e+107	2.058281e-09
29	29.0	71.0	2.883681e+41	1.986507e+66	5.728452e+107	6.185058e-09
30	30.0	70.0	3.162437e+42	5.223777e+65	1.651987e+108	1.783664e-08
31	31.0	69.0	3.366465e+43	1.359347e+65	4.576195e+108	4.940956e-08
32	32.0	68.0	3.482188e+44	3.499812e+64	1.218700e+109	1.315841e-07
33	33.0	67.0	3.503292e+45	8.913378e+63	3.122616e+109	3.371515e-07
34	34.0	66.0	3.431165e+46	2.245099e+63	7.703305e+109	8.317323e-07
35	35.0	65.0	3.274312e+47	5.591567e+62	1.830854e+110	1.976788e-06
36	36.0	64.0	3.046929e+48	1.376712e+62	4.194743e+110	4.529099e-06
37	37.0	63.0	2.766941e+49	3.350173e+61	9.269731e+110	1.000861e-05
38	38.0	62.0	2.453840e+50	8.055759e+60	1.976754e+111	2.134318e-05
39	39.0	61.0	2.126661e+51	1.913629e+60	4.069640e+111	4.394024e-05
40	40.0	60.0	1.802345e+52	4.489667e+59	8.091931e+111	8.736925e-05
41	41.0	59.0	1.494628e+53	1.040077e+59	1.554529e+112	1.678437e-04
42	42.0	58.0	1.213495e+54	2.378471e+58	2.886264e+112	3.116324e-04
43	43.0	57.0	9.651522e+54	5.367756e+57	5.180702e+112	5.593647e-04
44	44.0	56.0	7.523800e+55	1.195165e+57	8.992179e+112	9.708931e-04
45	45.0	55.0	5.751527e+56	2.624675e+56	1.509589e+113	1.629916e-03
46	46.0	54.0	4.313645e+57	5.683351e+55	2.451596e+113	2.647009e-03
47	47.0	53.0	3.175577e+58	1.213047e+55	3.852126e+113	4.159172e-03
48	48.0	52.0	2.295678e+59	2.551250e+54	5.856849e+113	6.323689e-03
49	49.0	51.0	1.630400e+60	5.285459e+53	8.617411e+113	9.304290e-03
50	50.0	50.0	1.138019e+61	1.078234e+53	1.227050e+114	1.324857e-02
51	51.0	49.0	7.809934e+61	2.165128e+52	1.690951e+114	1.825733e-02
52	52.0	48.0	5.271706e+62	4.277873e+51	2.255169e+114	2.434925e-02
53	53.0	47.0	3.501208e+63	8.313276e+50	2.910651e+114	3.142654e-02
54	54.0	46.0	2.288753e+64	1.588309e+50	3.635246e+114	3.925006e-02
55	55.0	45.0	1.473125e+65	2.982131e+49	4.393050e+114	4.743213e-02
56	56.0	44.0	9.338557e+65	5.499832e+48	5.136049e+114	5.545435e-02
57	57.0	43.0	5.832502e+66	9.958543e+47	5.808323e+114	6.271295e-02
58	58.0	42.0	3.590006e+67	1.769493e+47	6.352491e+114	6.858837e-02
59	59.0	41.0	2.178343e+68	3.083764e+46	6.717494e+114	7.252935e-02
60	60.0	40.0	1.303375e+69	5.268097e+45	6.866305e+114	7.413607e-02

61	61.0	39.0	7.692049e+69	8.816899e+44	6.782001e+114	7.322584e-02
62	62.0	38.0	4.478757e+70	1.444786e+44	6.470846e+114	6.986626e-02
63	63.0	37.0	2.573508e+71	2.316534e+43	5.961620e+114	6.436811e-02
64	64.0	36.0	1.459662e+72	3.631855e+42	5.301279e+114	5.723836e-02
65	65.0	35.0	8.174106e+72	5.563692e+41	4.547821e+114	4.910320e-02
66	66.0	34.0	4.520528e+73	8.321762e+40	3.761876e+114	4.061729e-02
67	67.0	33.0	2.469423e+74	1.214334e+40	2.998705e+114	3.237727e-02
68	68.0	32.0	1.332762e+75	1.727286e+39	2.302061e+114	2.485555e-02
69	69.0	31.0	7.108064e+75	2.392777e+38	1.700801e+114	1.836369e-02
70	70.0	30.0	3.746965e+76	3.225047e+37	1.208414e+114	1.304735e-02
71	71.0	29.0	1.952644e+77	4.224953e+36	8.249828e+113	8.907409e-03
72	72.0	28.0	1.006154e+78	5.373844e+35	5.406914e+113	5.837890e-03
73	73.0	27.0	5.127250e+78	6.628530e+34	3.398613e+113	3.669511e-03
74	74.0	26.0	2.584411e+79	7.919040e+33	2.046606e+113	2.209737e-03
75	75.0	25.0	1.288760e+80	9.150890e+32	1.179330e+113	1.273332e-03
76	76.0	24.0	6.359012e+80	1.021305e+32	6.494489e+112	7.012154e-04
77	77.0	23.0	3.105180e+81	1.099162e+31	3.413096e+112	3.685148e-04
78	78.0	22.0	1.500837e+82	1.138771e+30	1.709110e+112	1.845341e-04
79	79.0	21.0	7.181220e+82	1.133619e+29	8.140765e+111	8.789652e-05
80	80.0	20.0	3.402103e+83	1.082091e+28	3.681383e+111	3.974821e-05
81	81.0	19.0	1.596048e+84	9.882105e+26	1.577232e+111	1.702950e-05
82	82.0	18.0	7.415785e+84	8.612844e+25	6.387100e+110	6.896206e-06
83	83.0	17.0	3.413048e+85	7.144295e+24	2.438382e+110	2.632742e-06
84	84.0	16.0	1.556187e+86	5.622825e+23	8.750169e+109	9.447631e-07
85	85.0	15.0	7.030306e+86	4.184428e+22	2.941781e+109	3.176265e-07
86	86.0	14.0	3.147288e+87	2.933010e+21	9.231027e+108	9.966817e-08
87	87.0	13.0	1.396383e+88	1.927800e+20	2.691947e+108	2.906518e-08
88	88.0	12.0	6.140911e+88	1.182141e+19	7.259426e+107	7.838063e-09
89	89.0	11.0	2.677161e+89	6.723080e+17	1.799877e+107	1.943342e-09
90	90.0	10.0	1.157129e+90	3.521613e+16	4.074959e+106	4.399768e-10
91	91.0	9.0	4.959123e+90	1.684982e+15	8.356034e+105	9.022080e-11
92	92.0	8.0	2.107627e+91	7.290789e+13	1.536627e+105	1.659109e-11
93	93.0	7.0	8.883762e+91	2.817696e+12	2.503174e+104	2.702698e-12
94	94.0	6.0	3.714169e+92	9.574696e+10	3.556203e+103	3.839663e-13
95	95.0	5.0	1.540403e+93	2.802350e+09	4.316747e+102	4.660828e-14
96	96.0	4.0	6.338115e+93	6.868505e+07	4.353337e+101	4.700335e-15
97	97.0	3.0	2.587519e+94	1.353400e+06	3.501948e+100	3.781083e-16
98	98.0	2.0	1.048209e+95	2.010000e+04	2.106901e+99	2.274838e-17
99	99.0	1.0	4.214013e+95	2.000000e+02	8.428026e+97	9.099810e-19
100	100.0	0.0	1.681391e+96	1.000000e+00	1.681391e+96	1.815412e-20

	S_a	S_b	S_total
0	0.000000	187.529022	187.529022
1	5.703782	186.433749	192.137531
2	10.717746	185.331775	196.049521
3	15.329560	184.223010	199.552571
4	19.656999	183.107362	202.764361

5	23.764589	181.984735	205.749323
6	27.693141	180.855032	208.548173
7	31.470816	179.718154	211.188970
8	35.118222	178.574000	213.692222
9	38.651097	177.422465	216.073562
10	42.081853	176.263444	218.345297
11	45.420530	175.096827	220.517357
12	48.675417	173.922503	222.597919
13	51.853470	172.740357	224.593828
14	54.960616	171.550273	226.510890
15	58.001959	170.352132	228.354091
16	60.981943	169.145809	230.127752
17	63.904472	167.931178	231.835650
18	66.773002	166.708112	233.481114
19	69.590614	165.476477	235.067091
20	72.360073	164.236136	236.596209
21	75.083872	162.986951	238.070823
22	77.764270	161.728778	239.493048
23	80.403328	160.461469	240.864797
24	83.002926	159.184874	242.187800
25	85.564794	157.898836	243.463630
26	88.090523	156.603196	244.693718
27	90.581583	155.297789	245.879372
28	93.039339	153.982447	247.021785
29	95.465057	152.656994	248.122050
30	97.859917	151.321252	249.181169
31	100.225022	149.975036	250.200058
32	102.561405	148.618155	251.179560
33	104.870032	147.250414	252.120446
34	107.151814	145.871610	253.023425
35	109.407607	144.481535	253.889143
36	111.638219	143.079974	254.718192
37	113.844412	141.666703	255.511115
38	116.026909	140.241493	256.268402
39	118.186393	138.804107	256.990500
40	120.323514	137.354299	257.677813
41	122.438887	135.891816	258.330703
42	124.533100	134.416393	258.949493
43	126.606711	132.927760	259.534471
44	128.660251	131.425634	260.085886
45	130.694231	129.909722	260.603953
46	132.709134	128.379721	261.088855
47	134.705425	126.835316	261.540741
48	136.683549	125.276179	261.959727
49	138.643931	123.701969	262.345900
50	140.586980	122.112334	262.699314
51	142.513087	120.506904	263.019991

52	144.422630	118.885296	263.307925
53	146.315969	117.247108	263.563077
54	148.193453	115.591924	263.785377
55	150.055417	113.919308	263.974724
56	151.902183	112.228802	264.130985
57	153.734062	110.519930	264.253992
58	155.551355	108.792193	264.343548
59	157.354351	107.045065	264.399416
60	159.143328	105.277998	264.421327
61	160.918559	103.490414	264.408973
62	162.680302	101.681705	264.362007
63	164.428812	99.851231	264.280043
64	166.164331	97.998317	264.162649
65	167.887098	96.122251	264.009349
66	169.597341	94.222278	263.819618
67	171.295281	92.297600	263.592881
68	172.981135	90.347370	263.328505
69	174.655112	88.370688	263.025800
70	176.317413	86.366596	262.684009
71	177.968236	84.334071	262.302308
72	179.607772	82.272022	261.879794
73	181.236207	80.179276	261.415483
74	182.853720	78.054578	260.908298
75	184.460488	75.896574	260.357062
76	186.056680	73.703804	259.760484
77	187.642464	71.474686	259.117150
78	189.218001	69.207503	258.425503
79	190.783447	66.900383	257.683829
80	192.338956	64.551277	256.890234
81	193.884679	62.157938	256.042617
82	195.420759	59.717882	255.138641
83	196.947339	57.228356	254.175695
84	198.464557	54.686291	253.150848
85	199.972548	52.088242	252.060790
86	201.471444	49.430316	250.901760
87	202.961373	46.708081	249.669455
88	204.442461	43.916444	248.358906
89	205.914830	41.049493	246.964323
90	207.378600	38.100281	245.478881
91	208.833887	35.060531	243.894419
92	210.280806	31.920218	242.201024
93	211.719469	28.666941	240.386409
94	213.149983	25.284975	238.434958
95	214.572457	21.753724	236.326182
96	215.986995	18.045042	234.032037
97	217.393698	14.118131	231.511829
98	218.792667	9.908475	228.701142

99	220.183999	5.298317	225.482317
100	221.567790	0.000000	221.567790

Plotting Entropy

```
[25]: fig2 = plt.figure()
      ax2 = fig2.add_subplot(111)

      ax2.plot(table2['q_A'], table2['S_a'], label='S_a')
      ax2.plot(table2['q_A'], table2['S_b'], label='S_b')
      ax2.plot(table2['q_A'], table2['S_total'], label='S_total')

      # labels
      ax2.set_ylabel('Entropy')
      ax2.set_xlabel('q_A')

      #legend
      ax2.legend()
```

Canvas(toolbar=Toolbar(toolitems=[('Home', 'Reset original view', 'home', ), ('Back', 'Back to previous ...

```
[25]: <matplotlib.legend.Legend at 0x7fac879a2890>
```

```
[ ]:
```